

Business recommendations and action plan

Based on the exploratory analysis of data and observed trends regarding no **-shows**, several operational recommendations can be formulated to help HealthConnect Clinic reduce the no-show rate and improve appointment management.

Cautionary note regarding interpretation: These recommendations are based on associations observed in the data. They do not, on their own, establish a causal relationship. The actual effects of the various actions will need to be evaluated after their implementation.

1. Reinforce reminders for appointments scheduled more than 30 days in advance

Observation

The time between booking and the appointment date appears to be one of the main factors associated with no-shows.

The no-show rate reaches **60.49%** for appointments booked more than 30 days in advance, compared to **27.81%** for those booked less than 7 days in advance.

Recommendation

Implement a **specific callback process for appointments with long booking lead times**.

This arrangement could include:

- Confirmation at the time of booking;
- An intermediate reminder before the appointment;
- A final reminder close to the date of the appointment;
- A simple possibility to confirm, cancel or reschedule the appointment.

Job objective

Reduce no-shows related to appointments scheduled well in advance and improve the use of available time slots.

Monitoring indicators

- No-show rate for appointments booked more than 30 days in advance;
- Confirmation rate;

Priority

High

2. Implement personalized follow-up for patients with repeated no-shows

Observation

The no-show rate increases progressively with the number of previous no-shows:

No-Shows previous	No-Show Rate observed
0	43,51 %
1	53,49 %
2	59,36 %
3+	68,82 %

This trend indicates that behavioral history is an important indicator for identifying appointments that require special attention.

Recommendation

Implement **patient segmentation based on No-Show history** and strengthen follow-up for patients with repeated history.

The possible actions are:

- personalized reminder;
- proactive appointment confirmation;
- telephone contact for the highest risk profiles;
- facilitation of rescheduling;
- monitoring of behavioral changes after intervention.

Job objective

Take preventative action with patients who have a history of missed appointments.

Monitoring indicators

- No-show rate of patients with 1+, 2+ and 3+ previous no-shows;
- Confirmation rate of at-risk patients;

Priority

High

3. Pay particular attention to patients living more than 20 km from the clinic

Observation

The observed no-show rate increases with distance.

It reaches **57.76%** for patients located more than 20 km from the clinic, compared to approximately **46.45%** for those located 5 km or less.

Recommendation

Identify patients living more than 20 km away as a **segment requiring special operational attention**.

The clinic could, in particular:

- Confirm appointments earlier;
- Strengthen reminders;
- Facilitate cancellations and rescheduling;
- Explore, when medically and operationally feasible, alternatives such as teleconsultation.

Job objective

Identify and reduce any potential obstacles related to geographical accessibility.

Monitoring indicators

- No-show rate for patients >20 km;
- Attendance rate per distance segment;
- Cancellation/re-scheduling rate;
- Use of alternative solutions.

Priority

Medium to high

Caution: Distance is associated with No-Show in the data, but it should not be considered a direct cause without further analysis.

4. Test and optimize the callback strategy per channel

Observation

No-show rates vary depending on status and callback channel:

Channel / status	Rate of No-Show
No recall	51.39%
Reminder sent	47,36 %
SMS	45,75 %
Email	48,41 %
WhatsApp	49,77 %

SMS has the lowest observed No-Show rate among the channels studied.

Recommendation

Don't immediately conclude that SMS is necessarily the most effective channel. It's better to implement a **structured evaluation of the different reminder channels**.

The clinic could:

- 1. Standardize the timing of sending reminders;
- 2. Track the receipt and confirmation of messages;
- 3. Compare attendance rates by channel;
- 4. Analyze results on comparable populations;
- 5. Implement, where possible, a controlled test of the different strategies.

Job objective

Identify the communication strategy that will improve confirmation and attendance rates.

Monitoring indicators

- No-show rate per channel;
- Confirmation rate;
- Attendance rate;
- Rescheduling rate;
- Callback response rate.

Priority **High**

5. Prioritize appointments that combine several risk factors

Observation

The analysis shows that several factors can combine and lead to particularly high levels of No-Show.

The intersection between **booking time** and **No-Show history** shows particularly high rates in certain combinations, especially when the appointment is booked well in advance and the patient already has several previous No-Shows.

Recommendation

Develop an **operational segmentation of high-risk appointments** by combining the main factors identified:

- booking lead time; • number of previous no-shows; • distance from the clinic; • status and callback channel.

Appointments that meet several risk criteria could benefit from a higher level of monitoring.

Job objective

Focus administrative resources on appointments with the highest observed level of risk rather than applying exactly the same strategy to all patients.

Monitoring indicators

- No-show rate by risk segment; • Number of at-risk appointments identified; • Attendance rate after intervention; • Number of no-shows avoided; • Effectiveness of interventions by segment.

Priority Very high

Prioritization plan

Priority	Action	Main objective
1	Segmenting at-risk profiles: Effectively targeting interventions	
2	Strengthening of reminders >30 days	Reducing No-Shows Related to Long Delays

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	Patient follow-up with Repeated no-shows	Addressing recurring behaviors
4	Optimizing callback channels	Identify the most effective communication strategy
5	Caution for patients >20 km	Reduce any potential accessibility barriers

Monitoring and evaluation framework

In order to measure the effectiveness of the recommendations, HealthConnect Clinic should regularly monitor a limited set of indicators.

KPI principal

Overall No-Show Rate

KPI segmentation

- No-show rate according to booking lead time;
- No-show rate according to the number of previous no-shows;
- No-show rate according to distance;
- No-show rate according to the callback channel.

Operational KPIs

- Confirmation rate;
- Attendance rate;
- Cancellation rate;
- Rescheduling rate;
- Number of appointments identified as high-risk.

The objective should therefore not only be to monitor the evolution of the overall No-Show rate, but also to determine **which interventions actually improve attendance in the most at-risk segments.**

Conclusion: Job Description

The analysis shows that HealthConnect Clinic could adopt a **more targeted no-show management strategy**, rather than a uniform approach applied to all appointments.

The results particularly highlight the importance of **booking lead time** and **no-show history**. Combining these factors also helps identify profiles requiring closer attention.

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The operational priority should therefore be to **segment appointments according to their level of risk**, and then adapt reminder and follow-up actions accordingly.

Finally, recommendations relating to callback channels and other factors will need to be **tested and evaluated over time**, in order to distinguish simple associations observed in the data from interventions that actually have an impact on reducing no-shows.